Problem-Solving & Data

ANSWER KEY

Section 1: Rates, Proportions & Dimensional Analysis

1 A car travels at a constant speed of 66 feet per second. What is this speed in miles per hour? (1 mile $= 5,280$ feet)

a. 40 mph

b. 45 **mph** ✓

c. 50 mph

d. 55 mph

$\frac{66 \text{ ft}}{1 \text{ s}} \times \frac{3600 \text{ s}}{1 \text{ hr}} \times \frac{1 \text{ mi}}{5280 \text{ ft}} = \frac{66 \times 3600}{5280} = 45$ mph.

2 On a map, 2.5 centimeters represents an actual distance of 40 kilometers. If two cities are 6.5 centimeters apart on the map, what is the actual distance between them?

a. 96 km

b. 104 **km** ✓

c. 112 km

d. 120 km

Set up a proportion: $\frac{40}{2.5} = 16$ km per cm. For 6.5 cm: $6.5 \times 16 = 104$ km.

3 A water hose fills a 120-gallon tank in 15 minutes. At this rate, how many gallons of water does the hose pump in 2 hours?

a. 720

b. 840

c. 960 ✓

d. $1,080$

Rate is $\frac{120}{15} = 8$ gallons per minute. In 2 hours (120 minutes): $8 \times 120 = 960$ gallons.

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Section 2: Percentages & Successive Changes

4 The price of an item was originally 80. It was increased by 20% and then the new price was discounted by 10%. What is the final price of the item?

a. 86.40 ✓

b. 88.00

c. 90.20

d. 92.00

Price after increase: $80 \times 1.20 = 96$. Price after discount: $96 \times 0.90 = 86.40$ dollars.

5 A store manager discounts a winter jacket by 30%, making the sale price 84. What was the original price of the jacket?

a. 110

b. 115

c. 120 ✓

d. 125

Sale price represents 70% of original: $0.70P = 84 \Rightarrow P = \frac{84}{0.70} = 120$ dollars.

6 In a chemistry lab, a 400-milliliter solution is 15% acid. How many milliliters of pure water must be added to dilute the solution to 10% acid?

a. 150 mL

b. 200 **mL** ✓

c. 250 mL

d. 300 mL

Amount of pure acid is $400 \times 0.15 = 60$ mL. In the new diluted solution with volume $400 + w$: $\frac{60}{400 + w} = 0.10 \Rightarrow 60 = 40 + 0.10w \Rightarrow 0.10w = 20 \Rightarrow w = 200$ mL.

Section 3: Scatterplots & Lines of Best Fit

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7 The line of best fit for a scatterplot relating weekly study hours x to exam score y is given by $\hat{y} = 4.5x + 52$. What is the predicted exam score for a student who studies 8 hours?

a. 84

b. 86

c. 88 ✓

d. 90

Substitute $x = 8$: $\hat{y} = 4.5(8) + 52 = 36 + 52 = 88$.

8 For the model $\hat{y} = 3x + 25$, a student with $x = 6$ had an actual exam score of 47. What is the residual for this data point?

a. $+4$ ✓

b. -4

c. $+2$

d. -2

Predicted value: $\hat{y} = 3(6) + 25 = 43$. Residual is $\text{Actual} - \text{Predicted} = 47 - 43 = +4$.

9 In the linear regression equation $\hat{y} = -2.4x + 68$, where x is the temperature in degrees Celsius and y is the heating cost in dollars, what does the slope -2.4 represent?

a. **For each $1\,^\circ\text{C}$ increase in temperature, heating cost decreases by 2.40 dollars.** ✓

b. For each $1\,^\circ\text{C}$ increase in temperature, heating cost increases by 2.40 dollars.

c. The heating cost when temperature is $0\,^\circ\text{C}$ is 2.40 dollars.

d. The maximum possible temperature is $28.3\,^\circ\text{C}$.

The slope $m = -2.4$ indicates the rate of change: for every 1 unit increase in x, y changes by -2.4 (decreases by 2.40 dollars).

Section 4: Data Tables & Proportions

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10 A company surveys 200 employees regarding remote work. If 120 full-time employees and 80 part-time employees participated, and 65% of full-time and 40% of part-time employees favor remote work, what percentage of all surveyed employees favor remote work?

a. 52.5%

b. 55.0% ✓

c. 57.5%

d. 60.0%

Full-time favoring: $120 \times 0.65 = 78$. Part-time favoring: $80 \times 0.40 = 32$. Total favoring $= 78 + 32 = 110$. Percentage $= \frac{110}{200} = 55\%$.

11 The ratio of boys to girls in a math competition is $5 : 7$. If there are 35 boys, how many total students are in the competition?

a. 70

b. 84 ✓

c. 91

d. 98

Each ratio unit is $\frac{35}{5} = 7$ students. Total ratio units are $5 + 7 = 12$. Total students $= 12 \times 7 = 84$.

12 If x is inversely proportional to y, and $x = 12$ when $y = 6$, what is the value of y when $x = 8$?

a. 4

b. 9 ✓

c. 12

d. 16

Inverse variation: $x \cdot y = k \Rightarrow 12 \cdot 6 = 72$. When $x = 8$: $8y = 72 \Rightarrow y = 9$.

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